

UNORDERED SELECTIONS ACTIVITY

MATH 300

Example of a big general discrete math problem: An **down-right path** on a $n \times n$ grid is a sequence $2n$ steps going either down or right at each step starting at the upper-left corner and ending at the bottom-right corner. Two examples of a 3×3 grid is pictured below. How many different down-right paths are there on a $n \times n$ grid?

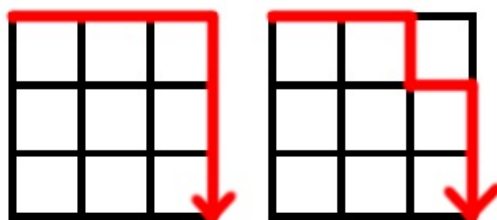


FIGURE 1. Two down-right paths on a 3×3 grid.

When facing a general problem like this, it's always a good idea to figure out what is going on for small n . That is $n = 1$, $n = 2$, $n = 3$ and so on. Draw pictures of the $n = 1$ and $n = 2$ cases.

Since there are six different solutions for $n = 2$, we might not want to continue to $n = 3$ with this process because there may be so many that it's hard to figure out repeated drawings.

Now that we have a good idea of the objects we are dealing with, it may be worth creating an easier model to work with, rather than drawing a bunch of picture. An important observation here is that every picture can be turned into a list of directions R for right and D for down. In this way the information in the above picture can be recored as RRRDDD and RRDRDD. That is we need a string consisting of 3 Rs and 3 Ds. Can we create a stage based process to create such strings?

Consider the two stage process to create a length 6 string with 3 Rs and 3 Ds.

- (1) From the 6 locations choose 3 of them to place Rs.
- (2) From the remaining 3 locations choose 3 of them to place Ds.

Questions:

- (1) In the first stage are we allowed to repeat choices of location?
- (2) In the first stage does the order of choosing locations matter? How can you check?
- (3) How many ways are there to do stage 1?
- (4) In the second stage why are we only choosing 3 locations?
- (5) In the second stage are we allowed to repeat choices of location?
- (6) In the second stage does the order of choosing locations matter?
- (7) How many ways are there to do stage 2?

Now consider strings of length 6 made with the following characters R,R,R,D,d, δ . Modify the above stage-based process to figure out how many strings can be created with those 6 characters. Finally can you answer the general question: How many different down-right paths are there on a $n \times n$ grid?